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Learning Objectives

- After completing this chapter, you will be able to:
- Explain the role of Big Data in modern business
- Describe the primary characteristics of Big Data and how these go beyond the traditional “3 Vs”
- Explain how the core components of the Hadoop framework operate
- Identify the major components of the Hadoop ecosystem
- Summarize the four major approaches of the NoSQL data model and how they differ from the relational model
- Describe the characteristics of NewSQL databases
- Understand how to work with document databases using MongoDB
- Understand how to work with graph databases using Neo4j

Big Data

Volume: the quantity of data to be stored

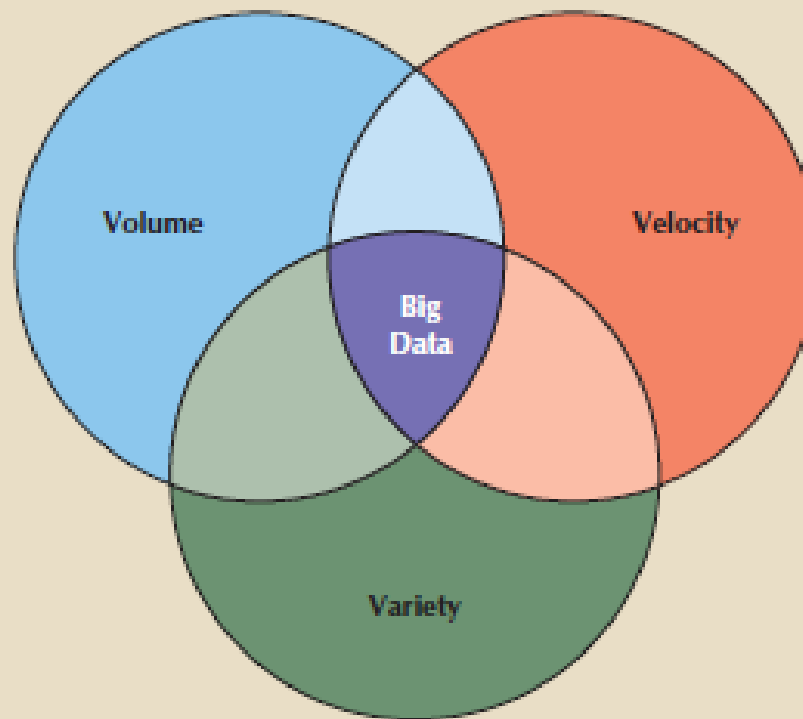
Velocity: the speed at which data is entering the system

Variety: the variations in the structure of the data to be stored

- That makes the data unsuitable for management by a relational database management system.

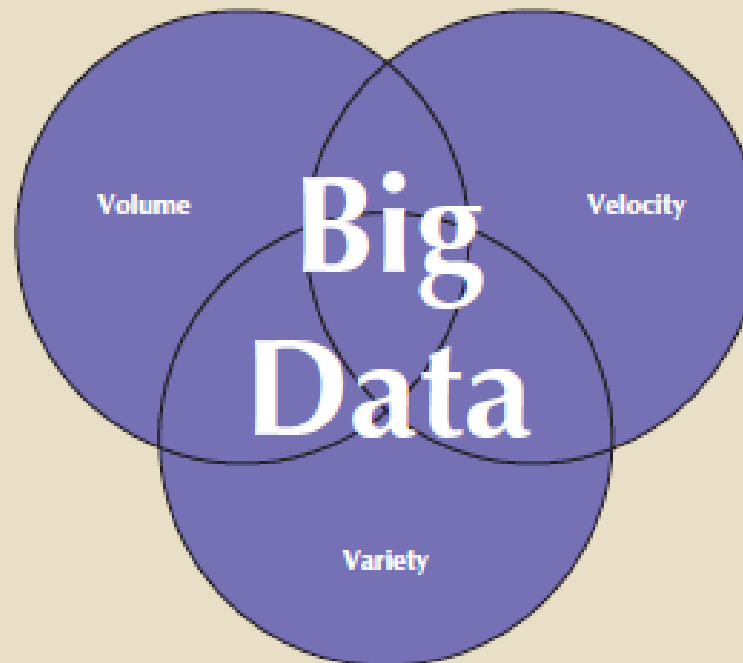
Big Data

FIGURE 14.1 ORIGINAL VIEW OF BIG DATA



Big Data

FIGURE 14.2 CURRENT VIEW OF BIG DATA



Big Data- Volume

- Volume, the quantity of data to be stored, is a key characteristic of Big Data.
- The storage capacities associated with Big Data are extremely large.
- as the quantity of data needing to be stored increases, the need for larger storage devices increases as well.
- **Scaling up** is keeping the same number of systems, but migrating each system to
- a larger system.
- **scaling out** means that when the workload exceeds the capacity of a server, the workload is spread out across a number of servers.
 - This is also referred to as clustering

Big Data- Velocity

- The rate at which new data enters the system as well as the rate at which the data must be processed.
- In addition for Big Data to be actionable, that data must be processed at a very rapid pace.
- The velocity of processing can be broken down into two categories.
 - Stream processing
 - Feedback loop processing

Big Data- Velocity

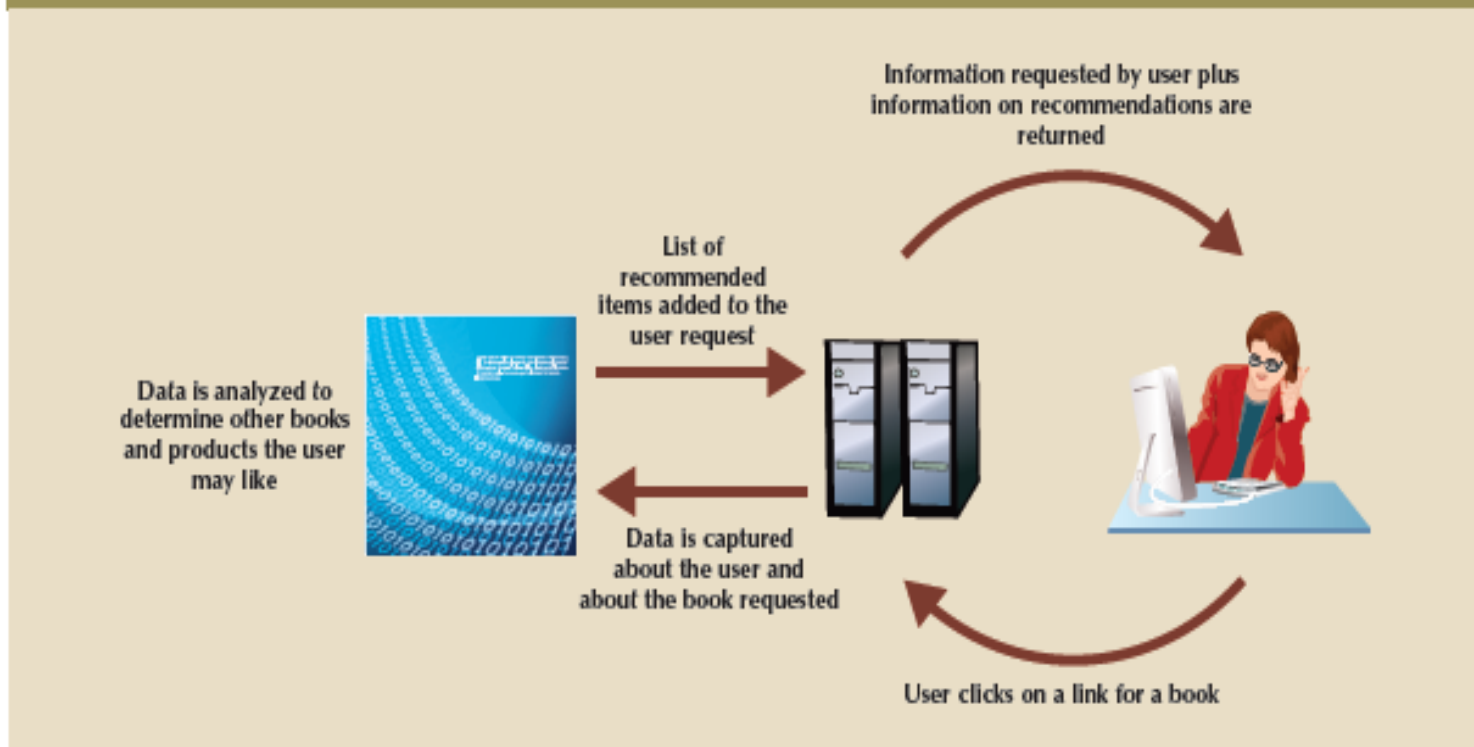
- **Stream processing** focuses on input processing, and it requires analysis of the data stream as it enters the system.
- In some situations, large volumes of data can enter the system at such a rapid pace that it is not feasible to try to store all of the data.
- The data must be processed and filtered as it enters the system to determine which data to keep and which data to discard.

Big Data- Velocity

- **Feedback loop processing** refers to the analysis of the data to produce actionable results.
- While stream processing could be thought of as focused on inputs, feedback loop processing can be thought of as focused on outputs.
- The process of capturing the data, processing it into usable information, and then acting on that information is a feedback loop.

Big Data- Velocity

FIGURE 14.3 FEEDBACK LOOP PROCESSING



Big Data- Variety

- Variety refers to the vast array of formats and structures in which the data may be captured.
- Data can be considered to be structured, unstructured, or semistructured.
- **Structured data** is data that has been organized to fit a predefined data model. Relational databases
- **Unstructured data** is data that is not organized to fit into a predefined data model.
 - maps, satellite images, emails, texts, tweets, videos, transcripts, and a whole host of other data forms.
- **Semistructured data** combines elements of both some parts of the data fit a predefined model while other parts do not.

Big Data- Other Characteristics

TABLE 14.2

ADDITIONAL Vs OF BIG DATA

CHARACTERISTIC	DESCRIPTION
Variability	Data meaning changes based on context.
Veracity	Data is correct.
Value (Viability)	Data can provide meaningful information.
Visualization	Data can be presented in such a way as to make it understandable.

Big Data- Other Characteristics

- **Variability** refers to the changes in the meaning of the data based on context.
- Variety is about differences in structure.
- Variability is about differences in meaning. Variability is especially relevant in areas such as sentiment analysis that attempt to understand the meanings of words

Big Data- Other Characteristics

- **Veracity** refers to the trustworthiness of the data.
- Can decision makers reasonably rely on the accuracy of the data and the information generated from it?
- This is especially pertinent given the automation of data capture and some of the analysis. Uncertainty about the data can arise from several causes,
 - having to capture only selected portions of the data due to high velocity.
 - customers' opinions and preferences can change over time, so comments at one point in time might not be suitable for action at another point in time.

Big Data- Other Characteristics

- **Value**, also called viability, refers to the degree to which the data can be analyzed to provide meaningful information that can add value to the organization.
- Just because a set of data can be captured does not mean that it should be captured.
- Only data that can form the basis for analysis that has the potential to impact organizational behavior should be included in a company's Big Data efforts.

Big Data- Other Characteristics

- **Visualization** is the ability to graphically present the data in such a way as to make it understandable.
- Volumes of data can leave decision makers awash in facts but with little understanding of what the facts mean.
- Visualization is a way of presenting the facts so that decision makers can comprehend the meaning of the information to gain insights.

Hadoop

- Big Data requires a different approach to distributed data storage that is designed for large-scale clusters.
- Hadoop has become the de facto standard for most Big Data storage and processing.
- Hadoop is a Java-based framework for distributing and processing very large data sets across clusters of computers.
 - Hadoop Distributed File System (HDFS) and
 - MapReduce.

Hadoop Distributed File System (HDFS)

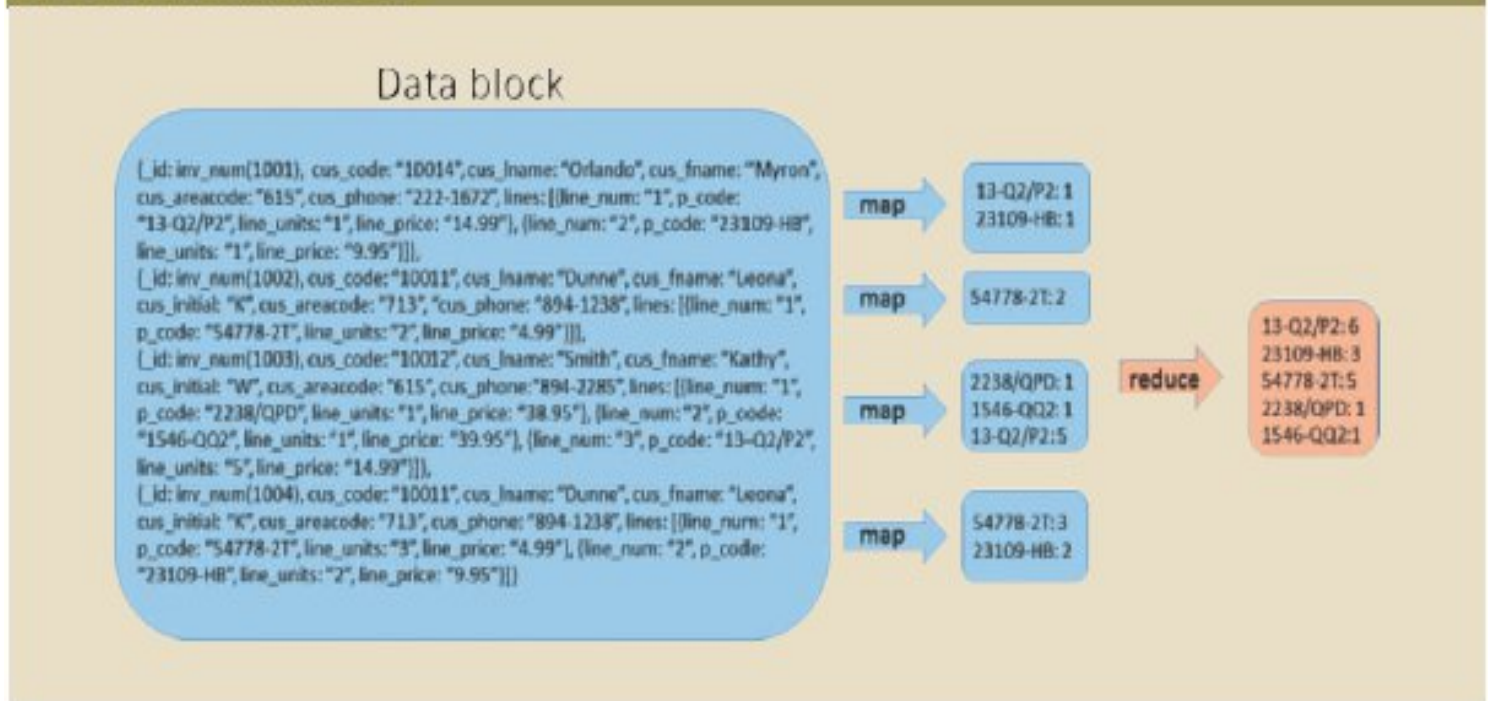
- The Hadoop Distributed File System (HDFS) approach to distributing data is based on several key assumptions:
 - **High volume**
 - **Write-once, read-many**
 - **Streaming access**
 - Hadoop is optimized for batch processing of entire files as a continuous stream of data.
 - **Fault tolerance**
 - Hadoop is designed to be distributed across thousands of low-cost, commodity computers.
 - HDFS is designed to replicate data across many different devices so that when one device fails, the data is still available from another device

MapReduce

- Is the computing framework used to process large data sets across clusters.
- MapReduce follows the principle of divide and conquer.
 - Takes a complex task,
 - Breaks it down into a collection of smaller subtasks,
 - Performs the subtasks all at the same time, and
 - Combines the result of each subtask to produce a final result for the original task.

MapReduce

FIGURE 14.5 MAPREDUCE



NoSQL

- NoSQL is array of nonrelational database technologies that have developed to address the challenges represented by Big Data.
- NoSQL could stand for “Not Only SQL.”
 - Keyvalue data stores,
 - Document databases,
 - Column-oriented databases, and
 - Graph databases

NoSQL

TABLE 14.3

NoSQL DATABASES

NoSQL CATEGORY	EXAMPLE DATABASES	DEVELOPER
Key-value database	Dynamo Riak Redis Voldemort	Amazon Basho Redis Labs LinkedIn
Document databases	MongoDB CouchDB OrientDB RavenDB	MongoDB, Inc. Apache OrientDB Ltd. Hibernate Rhinos
Column-oriented databases	HBase Cassandra Hypertable	Apache Apache (originally Facebook) Hypertable, Inc.
Graph databases	Neo4J ArangoDB GraphBase	Neo4j ArangoDB, LLC FactNexus

Key-Value Databases

- Stores data as a collection of key-value pairs.
- The key acts as an identifier for the value.
- The value can be anything such as text, an XML document, or an image.
- The database does not understand the contents of the value component or its meaning
- the database stores whatever value is provided for the key.
- It is the job of the applications that use the data to understand the meaning of the data in the value component

Key-Value Databases

- Key-value pairs are typically organized into “buckets.”
- A bucket can roughly be thought of as the KV database equivalent of a table.
- A bucket is a logical grouping of keys.
- Key values must be unique within a bucket, but they can be duplicated across buckets.
- All data operations are based on the bucket plus the key.

Key-Value Databases

FIGURE 14.7 KEY-VALUE DATABASE STORAGE

Bucket = Customer

Key	Value
10010	"LName Ramas FName Alfred Initial A Areacode 615 Phone 844-2573 Balance 0"
10011	"LName Dunne FName Leona Initial K Areacode 713 Phone 894-1238 Balance 0"
10014	"LName Orlando FName Myron Areacode 615 Phone 222-1672 Balance 0"

Document Databases

- Conceptually similar to key-value databases, and they can almost be considered a subtype of KV databases.
- A document database is a NoSQL database that stores data in tagged documents in key-value pairs.
- Unlike a KV database where the value component can contain any type of data, a document database always stores a document in the value component.
- The document can be in any encoded format, such as XML, JSON (JavaScript Object Notation), or BSON (Binary JSON).
- document databases attempt to understand the content of the value component.

- Document databases group documents into logical groups called **collections**.
- Document may be retrieved by specifying the collection and key.

FIGURE 14.8 DOCUMENT DATABASE TAGGED FORMAT

Collection = Customer

Key	Document
10010	{LName: "Ramas", FName: "Alfred", Initial: "A", Areacode: "615", Phone: "844-2573", Balance: "0"}
10011	{LName: "Dunne", FName: "Leona", Initial: "K", Areacode: "713", Phone: "894-1238", Balance: "0"}
10014	{LName: "Orlando", FName: "Myron", Areacode: "615", Phone: "222-1672", Balance: "0"}

Column-Oriented Databases

- column-oriented database or columnar database can refer to traditional, relational database technologies that use column-centric storage instead of row-centric storage.

Column-Oriented Databases

FIGURE 14.9 COMPARISON OF ROW-CENTRIC AND COLUMN-CENTRIC STORAGE

CUSTOMER relational table

Cus_Code	Cus_LName	Cus_FName	Cus_City	Cus_State
10010	Ramas	Alfred	Nashville	TN
10011	Dunne	Leona	Miami	FL
10012	Smith	Kathy	Boston	MA
10013	Olowski	Paul	Nashville	TN
10014	Orlando	Myron		
10015	O'Brian	Amy	Miami	FL
10016	Brown	James		
10017	Williams	George	Mobile	AL
10018	Farriss	Anne	Opp	AL
10019	Smith	Olette	Nashville	TN

Row-centric storage

Block 1 10010,Ramas,Alfred,Nashville,TN 10011,Dunne,Leona,Miami,FL	Block 4 10016,Brown,James,NULL,NULL 10017,Williams,George,Mobile,AL
Block 2 10012,Smith,Kathy,Boston,MA 10013,Olowski,Paul,Nashville,TN	Block 5 10018,Farriss,Anne,OPP,AL 10019,Smith,Olette,Nashville,TN
Block 3 10014,Orlando,Myron,NULL,NULL 10015,O'Brian,Amy,Miami,FL	

Column-centric storage

Block 1 10010,10011,10012,10013,10014 10015,10016,10017,10018,10019	Block 4 Nashville,Miami,Boston,Nashville,NULL Miami,NULL,Mobile,Opp,Nashville
Block 2 Ramas,Dunne,Smith,Olowski,Orlando O'Brian,Brown,Williams,Farriss,Smith	Block 5 TN,FL,MA,TN,NULL, FL,NULL,AL,AL,TN
Block 3 Alfred,Leona,Kathy,Paul,Myron Amy,James,George,Anne,Olette	

Graph Databases

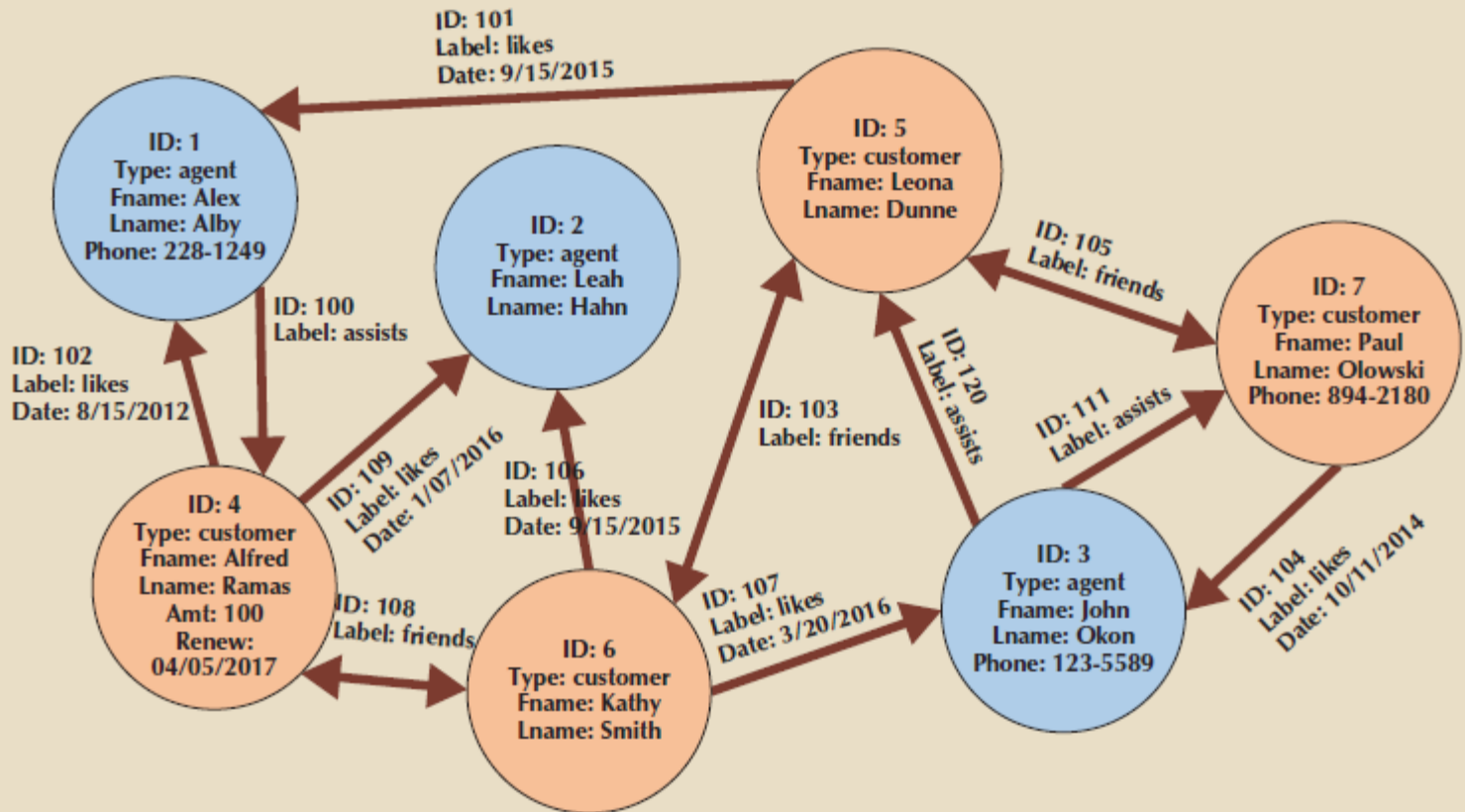
- A graph database is a NoSQL database based on graph theory to store data about relationship-rich environments.
- Graph theory is a mathematical and computer science field that models relationships, or edges, between objects called nodes.
- Modeling and storing data about relationships is the focus of graph databases

Graph Databases

- Interest in graph databases originated in the area of social networks.
- The primary components of graph databases are nodes, edges, and properties.
- The **node** is a specific instance of something we want to keep data about.
- An **edge** is a relationship between nodes.
- Note that edges can also have **properties**.

Graph Databases

FIGURE 14.11 GRAPH DATABASE REPRESENTATION



Working with Document Databases Using MongoDB

- A simple JSON document to store data on a book might look like this:
- `{_id: 101, title: 'Database Systems'}`
- This document contains two key:value pairs:
 - `_id` is a key with 101 as the associated value
 - `title` is a key with 'Database Systems' as the associated value
- `{_id: 101, title: 'Database Systems', author: ['Coronel', 'Morris']}`

Working with Document Databases Using MongoDB

```
{  
  _id: 101,  
  title: 'Database Systems',  
  author: ['Coronel', 'Morris']  
}
```

Working with Graph Databases Using Neo4j

- CREATE (:Member {mid: 1, fname: "Phillip", lname: "Stallings"})
- To retrieve our single member node.
- MATCH (m)
- RETURN(m)
- to retrieve Phillip Stallings
- MATCH (m {fname: "Phillip"}), (x {lname: "Stallings"})
- RETURN m